INNER joins

```
CREATE DATABASE IF NOT EXISTS college_dbms;
USE college_dbms;

DROP TABLE IF EXISTS Student;
DROP TABLE IF EXISTS Department;

CREATE TABLE Department (
   Dept_ID INT PRIMARY KEY,
   Dept_Name VARCHAR(50)
);

CREATE TABLE Student (
   Student_ID INT PRIMARY KEY,
   Name VARCHAR(50),
   Dept_ID INT,
   Marks INT,
   City VARCHAR(50)
);

INSERT INTO Department (Dept_ID, Dept_Name) VALUES
(10, 'CSE'),
(20, 'ECE'),
(30, 'MECH');

INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES
(101, 'Mohith Gowda MR', 10, 85, 'Delhi'),
(102, 'Vivek', 20, 92, 'Mumbai'),
(103, 'Rahul', 10, 76, 'Delhi'),
(104, 'Kiran', 30, 88, 'Bangalore'),
(105, 'Arjun', 20, 95, 'Delhi');

SELECT Student.Student_ID, Student.Name, Department.Dept_Name,
      Student.Marks
FROM Student
INNER JOIN Department
ON Student.Dept_ID = Department.Dept_ID;
```

	Student_ID	Name	Dept_Name	Marks
▶	101	Mohith Gowda MR	CSE	85
	102	Vivek	ECE	92
	103	Rahul	CSE	76
	104	Kiran	MECH	88
	105	Arjun	ECE	95

LEFT Join

```
CREATE TABLE Student (
    Student_ID INT PRIMARY KEY,
    Name VARCHAR(50),
    Dept_ID INT,
    Marks INT,
    City VARCHAR(50)
);

INSERT INTO Department (Dept_ID, Dept_Name) VALUES
(10, 'CSE'),
(20, 'ECE'),
(30, 'MECH');

INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES
(101, 'Mohith Gowda MR', 10, 85, 'Delhi'),
(102, 'Vivek', 20, 92, 'Mumbai'),
(103, 'Rahul', 10, 76, 'Delhi'),
(104, 'Kiran', 30, 88, 'Bangalore'),
(105, 'Arjun', 20, 95, 'Delhi');

SELECT Student.Name, Department.Dept_Name
FROM Student
LEFT JOIN Department
ON Student.Dept_ID = Department.Dept_ID;
```

	Name	Dept_Name
▶	Mohith Gowda MR	CSE
	Vivek	ECE
	Rahul	CSE
	Kiran	MECH
	Arjun	ECE

RIGHT Join

```
CREATE DATABASE IF NOT EXISTS college_dbms;
USE college_dbms;

DROP TABLE IF EXISTS Student;
DROP TABLE IF EXISTS Department;

CREATE TABLE Department (
    Dept_ID INT PRIMARY KEY,
    Dept_Name VARCHAR(50)
);

CREATE TABLE Student (
    Student_ID INT PRIMARY KEY,
    Name VARCHAR(50),
    Dept_ID INT,
    Marks INT,
    City VARCHAR(50)
);

INSERT INTO Department (Dept_ID, Dept_Name) VALUES
(10, 'CSE'),
(20, 'ECE'),
(30, 'MECH');

INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES
(101, 'Mohith Gowda MR', 10, 85, 'Delhi'),
(102, 'Vivek', 20, 92, 'Mumbai'),
(103, 'Rahul', 10, 76, 'Delhi'),
(104, 'Kiran', 30, 88, 'Bangalore'),
(105, 'Arjun', 20, 95, 'Delhi');

SELECT Student.Name, Department.Dept_Name
FROM Student
RIGHT JOIN Department
ON Student.Dept_ID = Department.Dept_ID;
```

	Name	Dept_Name
▶	Rahul	CSE
	Mohith Gowda MR	CSE
	Arjun	ECE
	Vivek	ECE
	Kiran	MECH

FULL OUTER Join

```
CREATE DATABASE IF NOT EXISTS college_dbms;
USE college_dbms;

DROP TABLE IF EXISTS Student;
DROP TABLE IF EXISTS Department;

CREATE TABLE Department (
    Dept_ID INT PRIMARY KEY,
    Dept_Name VARCHAR(50)
);

CREATE TABLE Student (
    Student_ID INT PRIMARY KEY,
    Name VARCHAR(50),
    Dept_ID INT,
    Marks INT,
    City VARCHAR(50)
);

INSERT INTO Department (Dept_ID, Dept_Name) VALUES
(10, 'CSE'),
(20, 'ECE'),
(30, 'MECH');

INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES
(101, 'Mohith Gowda MR', 10, 85, 'Delhi'),
(102, 'Vivek', 20, 92, 'Mumbai'),
(103, 'Rahul', 10, 76, 'Delhi'),
(104, 'Kiran', 30, 88, 'Bangalore'),
(105, 'Arjun', 20, 95, 'Delhi');

SELECT Student.Name, Department.Dept_Name
FROM Student
LEFT JOIN Department ON Student.Dept_ID = Department.Dept_ID

UNION

SELECT Student.Name, Department.Dept_Name
FROM Student
RIGHT JOIN Department ON Student.Dept_ID = Department.Dept_ID;
```

	Name	Dept_Name
▶	Mohith Gowda MR	CSE
	Vivek	ECE
	Rahul	CSE
	Kiran	MECH
	Arjun	ECE

https://github.com/mohithgowdamrbca25-spec/DBMS.git

AGGREGATE FUNCTION

GROUP BY

m1

Limit to 1000 rows

```
    Student_ID INT,
    Name VARCHAR(50),
    Class INT,
    Marks INT
);

INSERT INTO STUDENT (Student_ID, Name, Class, Marks)
VALUES
    (1, 'Mohith Gowda MR', 1, 85),
    (2, 'Rahul', 1, 65),
    (3, 'Arjun', 1, 75),
    (4, 'Karthik', 2, 80),
    (5, 'Vivek', 2, 65);

SELECT *
FROM STUDENT;

SELECT
    Class,
    COUNT(*) AS Total_Students,
    AVG(Marks) AS Average_Marks
FROM STUDENT
```

OUTPUT

	Class	Total_Students	Average_Marks
▶	10A	81	80.6667
	10B	54	83.0000

ORDER BY

m1

Limit to 1000 rows

```sql
CREATE TABLE STUDENT (
    Student_ID INT,
    Name VARCHAR(50),
    Class INT,
    Marks INT
);

INSERT INTO STUDENT (Student_ID, Name, Class, Marks)
VALUES
    (1, 'Mohith Gowda MR', 1, 85),
    (2, 'Rahul', 1, 65),
    (3, 'Arjun', 1, 75),
    (4, 'Karthik', 2, 80),
    (5, 'Vivek', 2, 65);

SELECT *
FROM STUDENT;

SELECT *
FROM STUDENT
ORDER BY Name ASC;
```

OUTPUT

Student_ID	Student_Name	Class	Marks	Fee_Paid	Date_of_Birth	Admission_Date
115	Mohith Gowda MR	10A	85	55000.00	2007-02-13	2023-06-10
116	Rahul	10A	92	53000.00	2007-11-20	2023-06-12
117	Arjun	10B	78	45000.00	2007-08-08	2023-06-11
118	Karthik	10B	88	46000.00	2007-01-03	2023-06-15
119	Vivek	10A	65	49000.00	2007-01-30	2023-06-13

https://github.com/mohithgowdamrbca25-spec/DBMS.git

Procedure

```
mysql> CREATE DATABASE HospitalDB;
Query OK, 1 row affected (0.01 sec)

mysql> USE HospitalDB;
Database changed
mysql>
mysql> CREATE TABLE Patient (
    ->     Patient_ID INT PRIMARY KEY,
    ->     Patient_Name VARCHAR(50),
    ->     Disease VARCHAR(50),
    ->     Age INT
    -> );
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> INSERT INTO Patient VALUES
    -> (301, 'Aarav', 'Fever', 21),
    -> (302, 'Diya', 'Cold', 19),
    -> (303, 'Rohan', 'Headache', 25),
    -> (304, 'Ananya', 'Cough', 22),
    -> (305, 'Kiran', 'Allergy', 24),
    -> (306, 'Meera', 'Fever', 20);
Query OK, 6 rows affected (0.01 sec)
Records: 6  Duplicates: 0  Warnings: 0

mysql>
mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE GetPatientDetails(IN p_patient_id INT)
    -> BEGIN
    ->     SELECT *
    ->     FROM Patient
    ->     WHERE Patient_ID = p_patient_id;
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql>
mysql> CALL GetPatientDetails(303);
```

OUTPUT:

```
mysql>
mysql> CALL GetPatientDetails(303);
+------------+--------------+---------+------+
| Patient_ID | Patient_Name | Disease | Age  |
+------------+--------------+---------+------+
|        303 | Rohan        | Headache |   25 |
+------------+--------------+---------+------+
1 row in set (0.01 sec)

Query OK, 0 rows affected (0.01 sec)

mysql>
```

https://github.com/mohithgowdamrbca25-spec/DBMS.git

TRIGGER IN SQL

1) AFTER INSERT Trigger :

```
mysql> CREATE TABLE Student (
    ->     student_id INT AUTO_INCREMENT PRIMARY KEY,
    ->     student_name VARCHAR(100) NOT NULL,
    ->     major VARCHAR(50)
    -> );
ERROR 1046 (3D000): No database selected
mysql> CREATE DATABASE CollegeDB;
Query OK, 1 row affected (0.01 sec)

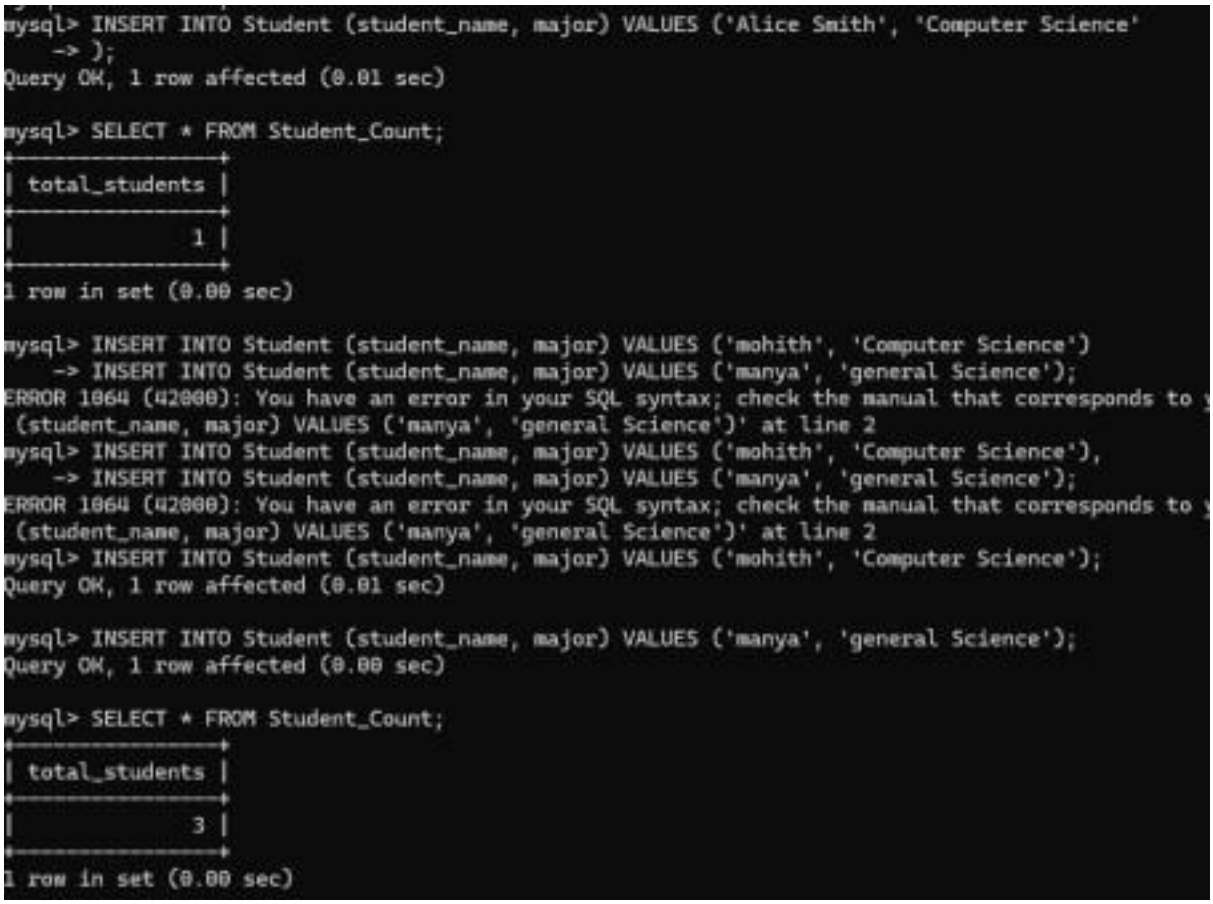
mysql> USE CollegeDB;
Database changed
mysql> CREATE TABLE Student (
    ->     student_id INT AUTO_INCREMENT PRIMARY KEY,
    ->     student_name VARCHAR(100) NOT NULL,
    ->     major VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> CREATE TABLE Student_Count (
    ->     total_students INT DEFAULT 0
    -> );
Query OK, 0 rows affected (0.02 sec)

mysql> INSERT INTO Student_Count (total_students) VALUES (0);
Query OK, 1 row affected (0.02 sec)

mysql> DELIMITER //
mysql>
mysql> CREATE TRIGGER after_student_insert
    -> AFTER INSERT ON Student
    -> FOR EACH ROW
    -> BEGIN
    ->     UPDATE Student_Count
    ->     SET total_students = total_students + 1;
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql> INSERT INTO Student (student_name, major) VALUES ('Alice Smith', 'Computer Science'
    -> );
```



2) AFTER UPDATE Trigger :

```
mysql> USE CompanyDB;
Database changed
mysql> CREATE TABLE Employee (
    ->     emp_id INT AUTO_INCREMENT PRIMARY KEY,
    ->     emp_name VARCHAR(100) NOT NULL,
    ->     salary DECIMAL(10, 2) NOT NULL);
Query OK, 0 rows affected (0.03 sec)

mysql> CREATE TABLE Salary_Audit (
    ->     audit_id INT AUTO_INCREMENT PRIMARY KEY,
    ->     emp_id INT,
    ->     old_salary DECIMAL(10, 2),
    ->     new_salary DECIMAL(10, 2),
    ->     change_time TIMESTAMP DEFAULT CURRENT_TIMESTAMP
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> DELIMITER //
mysql>
mysql> CREATE TRIGGER after_salary_update
    -> AFTER UPDATE ON Employee
    -> FOR EACH ROW
    -> BEGIN
    -> ^C
mysql> CREATE TRIGGER after_salary_update
    -> AFTER UPDATE ON Employee
    -> FOR EACH ROW
    -> BEGIN
    -> IF OLD.salary <> NEW.salary THEN
    ->         INSERT INTO Salary_Audit (emp_id, old_salary, new_salary)
    ->         VALUES (OLD.emp_id, OLD.salary, NEW.salary);
    ->     END IF;
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql> INSERT INTO Employee (emp_name, salary) VALUES ('John Doe', 50000.00);
```

```
mysql> INSERT INTO Employee (emp_name, salary) VALUES ('Mohith', 65000.00);
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO Employee (emp_name, salary) VALUES ('Manya', 58000.00);
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO Employee (emp_name, salary) VALUES ('Alice Smith', 45000.00);
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO Employee (emp_name, salary) VALUES ('Bob Jones', 72000.00);
Query OK, 1 row affected (0.01 sec)

mysql> UPDATE Employee SET salary = 75000.00 WHERE emp_name = 'Mohith';
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> UPDATE Employee SET salary = 62000.00 WHERE emp_name = 'Manya';
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> UPDATE Employee SET salary = 74500.00 WHERE emp_name = 'Shobi';
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0  Changed: 0  Warnings: 0

mysql> SELECT * FROM Salary_Audit;
+----------+--------+------------+------------+---------------------+
| audit_id | emp_id | old_salary | new_salary | change_time         |
+----------+--------+------------+------------+---------------------+
|        1 |      1 |   50000.00 |   55000.00 | 2026-09-29 12:02:36 |
|        2 |      2 |   65000.00 |   75000.00 | 2026-09-29 12:03:56 |
|        3 |      3 |   58000.00 |   62000.00 | 2026-09-29 12:04:09 |
+----------+--------+------------+------------+---------------------+
3 rows in set (0.00 sec)
```

3) AFTER DELETE Trigger :

```
mysql> USE HRDB;
Database changed
mysql> CREATE TABLE Employee (
    ->     emp_id INT AUTO_INCREMENT PRIMARY KEY,
    ->     emp_name VARCHAR(100) NOT NULL,
    ->     department VARCHAR(50) NOT NULL,
    ->     salary DECIMAL(10, 2) NOT NULL
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> CREATE TABLE Deleted_Employee_Log (
    ->     log_id INT AUTO_INCREMENT PRIMARY KEY,
    ->     emp_id INT,
    ->     emp_name VARCHAR(100),
    ->     department VARCHAR(50),
    ->     deleted_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> DELIMITER //
mysql>
mysql> CREATE TRIGGER prevent_management_deletion
    -> BEFORE DELETE ON Employee
    -> FOR EACH ROW
    -> BEGIN
    ->     -- Check if the employee belongs to Management
    ->     IF OLD.department = 'Management' THEN
    ->         SIGNAL SQLSTATE '45000'
    ->         SET MESSAGE_TEXT = 'Error: Employees in the Management department cannot be deleted!';
    ->     END IF;
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql> DELIMITER //
mysql>
mysql> CREATE TRIGGER log_deleted_employee
    -> AFTER DELETE ON Employee
    -> FOR EACH ROW
    -> BEGIN
```

```
    ->     INSERT INTO Deleted_Employee_Log (emp_id, emp_name, department)
    ->     VALUES (OLD.emp_id, OLD.emp_name, OLD.department);
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql> INSERT INTO Employee (emp_name, department, salary) VALUES ('Mohith', 'Management', 95000.00);
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO Employee (emp_name, department, salary) VALUES ('Alice Smith', 'IT', 60000.00);
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO Employee (emp_name, department, salary) VALUES ('Bob Jones', 'HR', 55000.00);
Query OK, 1 row affected (0.00 sec)

mysql> DELETE FROM Employee WHERE emp_name = 'Mohith';
ERROR 1644 (45000): Error: Employees in the Management department cannot be deleted!
mysql> DELETE FROM Employee WHERE emp_name = 'Alice Smith';
Query OK, 1 row affected (0.01 sec)

mysql> SELECT * FROM Employee;
+--------+-----------+------------+-----------+
| emp_id | emp_name  | department | salary    |
+--------+-----------+------------+-----------+
|      1 | Mohith    | Management | 95000.00 |
|      3 | Bob Jones | HR         | 55000.00 |
+--------+-----------+------------+-----------+
2 rows in set (0.00 sec)

mysql> SELECT * FROM Deleted_Employee_Log;
+--------+--------+-------------+------------+---------------------+
| log_id | emp_id | emp_name    | department | deleted_at          |
+--------+--------+-------------+------------+---------------------+
|      1 |      2 | Alice Smith | IT         | 2026-09-29 12:08:12 |
+--------+--------+-------------+------------+---------------------+
1 row in set (0.00 sec)
```